

Advanced Energy Industries, Inc.

Recommendation **HOLD** ★ ★ ★ ★ ★

Price
USD 87.83 [as of market close Oct 27, 2023] **12-Mo. Target Price** USD 114.00

Report Currency
USD

Investment Style
Mid-Cap Growth

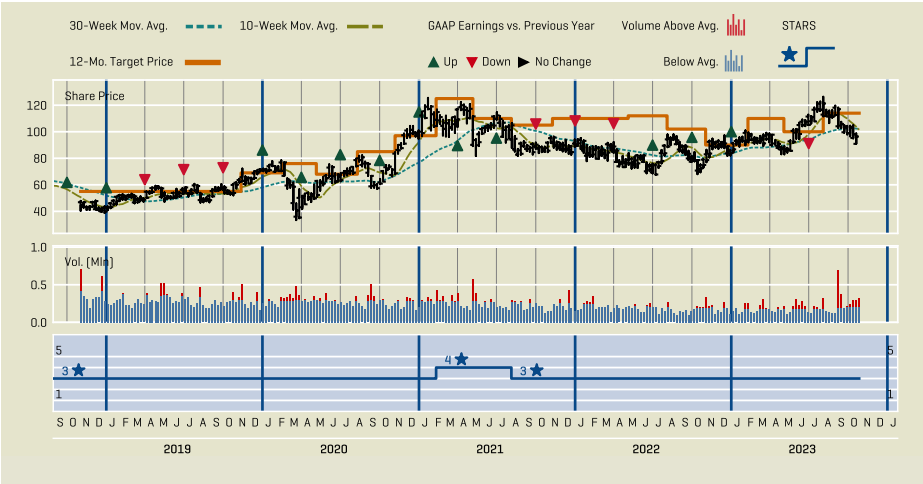
Equity Analyst Shreya Gheewala

GICS Sector Information Technology
Sub-Industry Electronic Equipment and Instruments

Summary Advanced Energy produces power conversion and control systems, as well as thermal and gas flow control systems used by semiconductor and industrial manufacturers.

Key Stock Statistics [Source: CFRA, S&P Global Market Intelligence (SPGMI), Company Reports]							
52-Wk Range	USD 126.38 - 75.95	Oper.EPS2023E	USD 4.74	Market Capitalization[B]	USD 3.31	Beta	1.59
Trailing 12-Month EPS	USD 6.17	Oper.EPS2024E	USD 5.99	Yield [%]	0.46	3-yr Proj. EPS CAGR[%]	18
Trailing 12-Month P/E	14.24	P/E on Oper.EPS2023E	18.53	Dividend Rate/Share	USD 0.4	SPGMI's Quality Ranking	B+
USD 10K Invested 5 Yrs Ago	20,099.0	Common Shares Outstg.[M]	38.00	Trailing 12-Month Dividend	USD 0.4	Institutional Ownership [%]	99.0

Price Performance



Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by **Shreya Gheewala** on Sep 01, 2023 07:00 PM ET, when the stock traded at **USD 111.28**.

Highlights

- ▶ Tight supply conditions hurt AEIS's ability to capitalize on record end market demand in 2022, but now some key markets have slowed coming into 2023 [e.g., Semiconductor Equipment, Data Center Computing, and Telecom/Networking]. We expect revenues to decline 8% in 2023 and rise 8% in 2024, assuming AEIS can work through extended lead times and chip away at an elevated backlog of \$645M exiting Q2 due to the downturn in the semi equipment market.
- ▶ We estimate AEIS will post a gross margin of 37%-38.5% for 2023-2024, vs. 37% in 2022, as supply chain constraints begin to subside, lowering input costs, new industrial and medical design wins come online. We forecast an EBIT margin of 12%-14.5% for 2023-2024, at continued compressed levels due to lower volumes and manufacturing ramps inside of 2023, down from 16.2% in 2022.
- ▶ AEIS's balance sheet is in a good spot, with \$455M in cash, net debt of \$11M, and low leverage [debt-to-EBITDA of 1.5x]. AEIS has no debt maturities in NTM. We advise investors to keep an eye on inventory turnover levels [2.7x in Q2 vs. 3.1x in Q1], within historical range, a possible sign AEIS is better sourcing raw materials to meet future demand.

Investment Rationale/Risk

- ▶ Following an extended period where AEIS's ability to source necessary components and capitalize on strong demand trends lagged other comparable companies, AEIS has recovered operating efficiency with a surge of design wins coming online to aid growth. We expect several more design wins in telecom and networking and industrial and medical markets, but our overall outlook is cautiously optimistic about management's ability to meet both near- and long-term targets. We positively view the completed acquisition of SL Power Electronics, which expanded AEIS's serviceable addressable market by ~\$400M to become a key player in medical power. We expect return on invested capital (ROIC) to expand to over 10% for the combined entity post synergies.
- ▶ Risks include weaker-than-expected growth in the global economy and/or a drop in semiconductor demand. An inability to meet long-term financial goals could also hurt the stock, in our view.
- ▶ Our 12-month target of \$114 is a P/E of 19x our 2024 EPS view, in line with the one-year average. Our conviction in AEIS's ability to meet three-year goals [established in Q4 2020] of \$7.50 in adj.-EPS and a 20%+ ROIC is reduced following recent operational missteps.

Analyst's Risk Assessment

LOW	MEDIUM	HIGH
-----	--------	------

Our risk assessment reflects the historical cyclicality of the semiconductor equipment industry, short lead times, high customer concentration, and the dynamic nature of changes in semiconductor technology. Risks are only partly countered by recent outperformance and design wins in power applications related to the Industrial and Medical end markets in the second half of 2022.

Revenue/Earnings Data

Revenue (Million USD)					
	1Q	2Q	3Q	4Q	Year
2024	E 430	E 452	E 471	E 485	E 1,838
2023	425	416	E 423	E 441	E 1,697
2022	397	441	516	491	1,845
2021	352	361	346	397	1,456
2020	315	340	390	371	1,416
2019	141	135	175	338	789

Earnings Per Share (USD)					
	1Q	2Q	3Q	4Q	Year
2024	E 1.22	E 1.38	E 1.60	E 1.79	E 5.99
2023	1.24	1.11	E 1.15	E 1.23	E 4.74
2022	1.24	1.44	2.12	1.70	6.49
2021	1.29	1.25	0.89	1.36	4.78
2020	0.91	1.18	1.66	1.49	5.23
2019	0.58	0.45	0.54	0.87	2.44

Fiscal Year ended Dec 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data

Amount [USD]	Date Decl.	Ex-Div. Date	Stk. of Record	Payment Date
0.1000	Jul 27	Aug 18	Aug 21	Sep 01 '23
0.1000	Apr 27	May 19	May 22	Jun 02 '23
0.1000	Feb 03	Feb 16	Feb 20	Mar 03 '23
0.1000	Nov 04	Nov 18	Nov 21	Dec 02 '22

Dividends have been paid since 2021 . Source: Company reports
Past performance is not an indication of future performance and should not be relied as such.

Forecasts are not a reliable indicator of future performance.
Dividends paid in currencies other than the Trading currency have been accordingly converted for display purposes.



Advanced Energy Industries, Inc.

Business Summary May 09, 2023

CORPORATE OVERVIEW. Advanced Energy Industries (AEIS) designs, manufactures, sells, and supports power conversion products that transform power into various usable forms. AEIS’s products are used in semiconductor and industrial thin film processes that contribute to the production of semiconductor devices, flat panel displays, digital storage and memory, architectural glass, industrial lasers, and medical devices.

AEIS’s thin-film deposition power conversion systems include direct current (DC), pulsed DC mid-frequency and radio frequency (RF) power supplies, matching networks, and RF instrumentation. The company’s power conversion systems refine, modify, and control the raw electrical power from a utility and convert it into power that may be customized and is predictable and repeatable. These power conversion systems are intended to be used by semiconductor, flat panel, and other similar thin-film manufacturers. AEIS’s thermal instrumentation products, used in the semiconductor industry and light-emitting diode (LED) industries, provide temperature-measurement solutions. These products are used in rapid thermal processing, chemical vapor deposition, and other semiconductor and solar applications requiring noncontact temperature measurement.

The company’s global support services group offers in-warranty and out-of-warranty repair services in the regions in which it operates. Semiconductor device manufacturers, specifically, have become more sensitive to the high costs of system downtime, causing them to require suppliers to offer comprehensive repair service and customer support.

Primary competitors include MKS Instruments, Comdel, Daihen Kyosan, Huttinger Electronics, SMA America, Inc., Schneider Electronics, Siemens, SatCon, STEC, and Brooks Automation. Although the company serves a variety of different markets, the global economy greatly influences demand, sales, and profitability. We see a number of AEIS’s primary segments as cyclical, but with secular growth underpinnings. Examples include semiconductor equipment, flat panel displays, data storage, and solar cells.

MARKET PROFILE. The semiconductor market is being driven by the rapid adoption of solid-state drives (SSD), deploying the latest 3D-NAND memory devices and a ramp of advanced logic devices at the 10 nm and below technology node. The industry’s transition to 3D memory devices and advanced Logic is generating increasing demand for RF power supplies, matches, and accessories.

The growing number of steps associated with the deposition and etch processes is driving an increase in the number of process chambers per fab and higher content of more advanced power solutions per chamber. As etching processes become more challenging due to increasing aspect ratios in advanced 3D devices, more advanced RF technology that includes pulsing and increased control and instrumentation is needed.

Following record levels of demand in the first half of 2018, the Semiconductor Equipment market experienced a decline, and, as a result, Advanced Energy’s revenue from the Semiconductor Equipment market declined 24.5% during 2019, as compared to 2018. Although the Semiconductor Equipment market improved during the fourth quarter of 2019 and throughout 2020, the company believes it is unclear how long this recovery will last.

FINANCIAL TRENDS. AEIS, along with most semiconductor equipment peers, has historically been highly cyclical, with periods of strong growth and high margins leading to over-investment and excess supply. Going forward, we think AEIS will experience less cyclicity due to the increased size and maturation of the semiconductor industry, rising diversity among end-users, both geographically and by end-products (e.g., life sciences and industrials).

The company sells products and services to approximately 370 original equipment manufacturers (OEMs) and integrators and directly to more than 1,500 end users. AEIS’s top ten largest customers accounted for approximately 58.3% of sales in 2021 [most recent data] with Applied Materials Inc., its largest customer, accounting for 20.4% of sales in 2021, while Lam Research accounted for 10.1% in 2021.

AEIS operates under four product lines: Semiconductor Capital Equipment (50% of 2022 sales), Industrial and Medical (25%), Data Center Computing (18%), and Telecom and Networking (7%). Total sales increased 23.3% in 2022, driven by a 30.5% increase in semiconductor equipment revenue. The gross margin was flat at 36.6% in 2022, driven by a cross current of higher volumes and higher input costs related to supply constraints.

AEIS’s balance sheet remains healthy. At the end of 2022, AEIS had an average cash conversion cycle of 109.6 days—down from 114.5 days in 2021 and lower than the company’s 10-year average of 123 days. AEIS’s backlog was \$850 million as of December 31, 2022, due to supply chain constraints, a boost in demand from Data Center Computing, Telecom Networking, and improvement in semiconductor equipment.

Corporate information

Investor contact N/A [970 221 0108]	
Office 1595 Wynkoop Street, Suite 800, Denver, Colorado, 80202	
Telephone 970 221 0108	
Fax N/A	
Website www.advancedenergy.com	
Officers	
Executive VP, General Counsel & Corporate Secretary E. K. Vonne	Executive VP & CFO P. R. Oldham
Executive VP & COO E. B. Acebedo	Senior VP & CTO R. Heckman
President, CEO & Director S. D. Kelley	Independent Chairman of the Board G. H. Beard
Board Members	
A. T. DelSanto	J. A. Roush
B. M. Shirley	L. T. Minnix
D. W. Reed	R. C. Foster
F. A. Ball	S. D. Kelley
G. H. Beard	T. M. Donikowski
Domicile Delaware	Auditor Ernst & Young LLP
Founded 1981	
Employees 12,000	
Stockholders 258	



Advanced Energy Industries, Inc.

Quantitative Evaluations					
Fair Value Rank	1	2	3	4	5
	LOWEST				HIGHEST
	Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued (1) to most undervalued (5).				
Fair Value Calculation	USD 83.51	Analysis of the stock's current worth, based on CFRA's proprietary quantitative model suggests that AEIS is overvalued by USD 4.32 or 4.92%			
Volatility	LOW		AVERAGE		HIGH
Technical Evaluation	NEUTRAL	Since September, 2023, the technical indicators for AEIS have been NEUTRAL"			
Insider Activity	UNFAVORABLE		NEUTRAL	FAVORABLE	

Expanded Ratio Analysis				
	2022	2021	2020	2019
Price/Sales	1.75	2.40	2.64	3.47
Price/EBITDA	10.60	15.94	15.77	32.03
Price/Pretax Income	13.39	23.49	23.63	40.79
P/E Ratio	13.22	19.05	18.54	29.18
Avg. Diluted Shares Outstg. [M]	37.72	38.36	38.54	38.50
Figures based on fiscal year-end price				
Key Growth Rates and Averages				
Past Growth Rate [%]		1 Year	3 Years	5 Years
Net Income		48.19	45.41	7.69
Sales		26.75	32.74	22.43
Ratio Analysis (Annual Avg.)				
Net Margin [%]		10.82	9.86	11.65
% LT Debt to Capitalization		22.78	24.75	20.56
Return on Equity [%]		20.84	18.31	17.96

Company Financials Fiscal year ending Dec 31										
Per Share Data [USD]	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
Tangible Book Value	15.90	13.28	11.38	7.55	11.79	10.94	8.11	4.70	9.62	7.09
Free Cash Flow	3.33	2.92	4.30	0.61	3.35	4.37	2.83	2.47	1.82	0.83
Earnings	5.35	3.51	3.51	1.47	3.74	3.39	2.92	2.03	1.69	1.47
Earnings [Normalized]	6.49	4.78	5.23	2.44	4.37	4.77	3.11	2.17	1.73	1.85
Dividends	0.40	0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Payout Ratio [%]	7.61	11.42	NM	NM	NM	NM	NM	NM	NM	NM
Prices: High	98.07	125.55	104.43	72.26	77.99	95.00	57.29	30.04	29.15	26.55
Prices: Low	67.55	81.71	33.38	40.76	38.74	53.79	24.25	21.12	16.00	13.48
P/E Ratio: High	15.10	26.30	20.00	29.60	17.80	19.90	18.40	13.80	16.80	14.40
P/E Ratio: Low	10.40	17.10	6.40	16.70	8.90	11.30	7.80	9.70	9.20	7.30
Income Statement Analysis [Million USD]										
Revenue	1,845	1,456	1,416	789.00	719.00	671.00	484.00	415.00	367.00	299.00
Operating Income	245.00	166.00	189.00	59.00	176.00	201.00	127.00	107.00	88.00	52.00
Depreciation + Amortization	60.00	53.00	48.00	26.00	14.00	9.00	8.00	9.00	10.00	7.00
Interest Expense	7.00	5.00	6.00	3.00	0.00	0.00	0.00	1.00	0.00	N/A
Pretax Income	242.00	149.00	158.00	67.00	172.00	198.00	128.00	105.00	86.00	48.00
Effective Tax Rate	16.50	9.40	14.50	15.90	14.60	31.30	8.70	20.80	19.20	-23.60
Net Income	200.00	135.00	135.00	65.00	147.00	138.00	127.00	NM	47.00	32.00
Net Income [Normalized]	150.70	101.50	107.80	36.30	110.60	123.90	80.20	65.40	55.00	33.20
Balance Sheet and Other Financial Data [Million USD]										
Cash	461.00	547.00	483.00	349.00	352.00	410.00	287.00	170.00	124.00	150.00
Current Assets	1,189	1,162	980.00	867.00	568.00	595.00	438.00	322.00	377.00	417.00
Total Assets	1,992	1,817	1,648	1,532	816.50	733.30	571.50	462.50	684.60	653.00
Current Liabilities	393.00	370.00	296.00	320.00	110.00	106.00	95.00	104.00	113.00	122.00
Long Term Debt	353.00	373.00	305.00	322.00	N/A	N/A	N/A	N/A	N/A	N/A
Total Capital	1,551	1,375	1,250	1,125	607.00	521.00	392.00	264.00	475.00	478.00
Capital Expenditures	59.00	29.00	36.00	25.00	20.00	9.00	7.00	4.00	4.00	3.00
Cash from Operations	184.00	140.00	201.00	48.00	151.00	183.00	119.00	105.00	77.00	35.00
Current Ratio	3.02	3.14	3.31	2.71	5.15	5.61	4.61	3.10	3.35	3.41
% Long Term Debt of Capitalization	22.80	27.10	24.40	28.60	N/A	N/A	N/A	N/A	N/A	N/A
% Net Income of Revenue	10.80	9.30	9.50	8.20	20.50	20.50	26.30	-38.20	12.80	10.70
% Return on Assets	8.04	6.00	7.44	3.16	14.18	19.23	15.34	11.64	8.19	5.46
% Return on Equity	20.80	16.00	18.10	8.80	26.10	29.80	35.70	22.60	14.80	14.00

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

Advanced Energy Industries, Inc.

Sub-Industry Outlook

We have a neutral fundamental outlook for the electronic equipment and instruments sub-industry. Electronic equipment and instrument products range from thermal, radiation, and water measurement industrial hardware to mobile computing, data capture, and real-time location systems.

Products and solutions are sold through distributors, which strive to properly align customers with the needed hardware, software, and services. End markets serviced include (but are not limited to) retail, transportation, manufacturing, health care, warehouse and distribution, energy, and utilities.

The Manufacturing Purchasing Managers Index (PMI), a good gauge of health for the Component Distributor industry, stood at 49% in September 2023, which is a 1.4% percentage point increase from the August reading of 47.6%. This figure marks the first expansion after nine months of contraction, following a 30-month period of expansion. The new orders and backlog indexes contracted, while the customer inventories index indicated improved supply chain efficiency as output improved and customer inventories continued to decline. The new orders index printed 49.2% in September, 2.4% above the August reading. We expect new orders and PMI to start to slowly creep into expansion territory through 2023 and into 2024.

Across the sub-industry, we expect revenues to be flat in 2023, as benefits from diverse set of end market growth stories but is weighed on by macroeconomic headwinds. Both lingering component shortages and slowdown in end market demand will likely remain industry headwinds for the second half of 2023, but we expect progressive improvement in the supply chain and factory utilization to help neutralize overall effects, as new orders slowdown but many participants begin to chip away at elevated backlogs.

Longer term, we will be keeping an eye on prior areas of strength, such as test and instrument. Our hunch is that test and instruments benefits

greatly from increasing complexity of equipment sub-systems and printed circuit boards (PCBs), partially offset by softening demand for consumer electronic products. We expect qualifications in more nascent areas, like big data/artificial intelligence, to help pick up the slack and provide a boost in overall demand.

Input costs (commodities and raw materials) remain volatile and, as a result, present margin and mix pressure in the first half of 2023, but are alleviating slightly compared to the second half of 2022. Similarly, some pandemic-related cost overruns (e.g., freight expenses) could restrict profitability, but we think issues have a high likelihood of being largely contained exiting 2023. Companies continue to implement tight cost-control measures, leading to greater earnings leverage to help offset industry-wide headwinds.

Across the sub-industry, the average cash conversion cycle [the time it takes a company to turn inventory into cash] was around 60 days in first half 2023, slightly above the three-year average of 94 days. Averages also aligned with five-year historical averages, but were slightly aided by improved cash collections and lower inventories outstanding from the large recovery in demand, and potential double ordering from select customers.

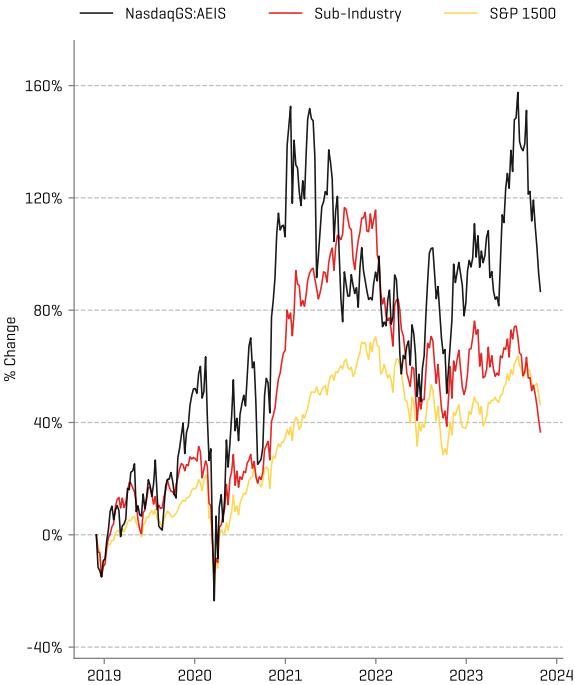
The S&P Electronic Equipment & Instruments Index decreased 1.4% year-to-date through October 6, underperforming the S&P 1500 Composite Index, which rose 11.2%. In 2022, the sub-industry index declined 29.4%, which was worse than the 19.1% contraction in the S&P 1500.

/ Shreya Gheewala

Industry Performance

GICS Sector: Information Technology
Sub-Industry: Electronic Equipment and Instruments

Based on S&P 1500 Indexes
Five-Year market price performance through Oct 28, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.
All Sector & Sub-Industry information is based on the Global Industry Classification Standard (GICS).
Past performance is not an indication of future performance and should not be relied upon as such.
Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Electronic Equipment and Instruments Peer Group*: Electronic Equipment and Instruments												
Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. (M)	30-Day Price Chg. (%)	1-Year Price Chg. (%)	P/E Ratio	Fair Value Calc.	Yield (%)	Return on Equity (%)	LTD to Cap (%)
Advanced Energy Industries, Inc.	AEIS	NasdaqGS	USD	88.40	3,328.0	-10.4	16.3	14.0	83.51	0.5	17.5	21.6
Arlo Technologies, Inc.	ARLO	NYSE	USD	8.12	764.0	-21.2	55.3	NM	N/A	N/A	-63.2	N/A
Badger Meter, Inc.	BMI	NYSE	USD	137.63	4,039.0	-8.5	27.4	48.0	N/A	0.8	18.7	N/A
Crane NXT, Co.	CXT	NYSE	USD	51.82	2,943.0	-6.4	N/A	15.0	N/A	1.1	N/A	47.5
Hollysys Automation Technologies Ltd.	HOLI	NasdaqGS	USD	20.24	1,256.0	6.8	31.5	12.0	N/A	N/A	9.1	1.4
Itron, Inc.	ITRI	NasdaqGS	USD	57.73	2,624.0	-2.9	22.0	28.0	45.55	N/A	3.3	26.4
Mirion Technologies, Inc.	MIR	NYSE	USD	6.87	1,368.0	-9.1	-13.5	NM	N/A	N/A	-17.5	29.7
Novanta Inc.	NOVT	NasdaqGS	USD	129.10	4,623.0	-7.4	-4.9	60.0	N/A	N/A	13.3	37.1
OSI Systems, Inc.	OSIS	NasdaqGS	USD	114.01	1,915.0	0.9	45.4	21.0	N/A	N/A	13.5	12.2
PAR Technology Corporation	PAR	NYSE	USD	30.08	826.0	-21.4	5.8	NM	N/A	N/A	-19.1	51.1
Vontier Corporation	VNT	NYSE	USD	29.25	4,526.0	-2.0	56.3	15.0	N/A	0.3	49.5	76.2

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.
NA-Not Available; NM-Not Meaningful.
Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

Advanced Energy Industries, Inc.

Analyst Research Notes and other Company News

August 04, 2023

04:20 PM ET... CFRA Reiterates Hold on Shares of Advanced Energy Industries, Inc. [AEIS 112.90***]:

We raise our 12-month price target by \$14 to \$114 on a forward P/E of 19x our 2024 EPS estimate, in line with AEIS's one-year average. We lift our 2023 EPS estimate by \$0.07 to \$4.74 and keep 2024's as \$5.99. Q2 sales fell 6% Y/Y on cyclical weakness in semiconductor, partially offset by Industrial & Medical strength. Gross margin decreased 140 bps Y/Y on unfavorable product mix. By segment: Semiconductor Equipment [42% of Q2 sales] -24% Y/Y, Industrial & Medical [31%] +22% Y/Y, Data Center Computing [14%] -15% Y/Y, and Telecom & Networking [13%] +46% Y/Y. AEIS posted Q2 EPS of \$1.11, beating consensus by \$0.06. We view the launched next generation plasma power platforms, the eVerest and eVOSTM ME, favorably for growth and expect AEIS's innovative product designs to drive strong wins in 2023. That said, we keep our rating at Hold as we believe shares are close to fair value. / Emily Nasseff Mitsch

May 04, 2023

09:25 AM ET... CFRA Maintains Hold Opinion on Shares of Advanced Energy Industries, Inc. [AEIS 87.46***]:

We lower our 12-month target by \$10 to \$100 on a P/E of 16.7x our 2024 EPS view, near peers. We cut our 2023 EPS view to \$4.67 from \$4.69 and 2024's to \$5.99 from \$6.10. AEIS prints non-GAAP Q1 EPS of \$1.24 vs. \$1.24, above the \$1.14 consensus. AEIS realized 7% revenue growth Y/Y in Q1 to \$425M, driven by 48% growth in Industrial and Medical revenue [+3% Q/Q] to \$123M, with a sequential decline of 14% in overall backlog to \$756M, which we view positively. Telecom and Networking posted 36% growth Y/Y [+8% Q/Q] to \$48M, as supply constraints began to ease. This was offset by a 22% decline in Data Center Computing revenue from the prior year [-36%] to \$60M, due largely to component shortages and a softening of demand from hyperscale customers. Semiconductor Equipment also contracted 4% [-16%], in line with our expectations, to \$194M. We are encouraged by high-voltage business related to ion implant applications and design wins in etch/deposition applications in 2H 2023 and 2024, when semi spending normalizes. / Keven Young

February 09, 2023

10:35 AM ET... CFRA Maintains Hold Opinion on Shares of Advanced Energy Industries, Inc. [AEIS 97.20***]:

We raise our 12-month target by \$20 to \$110 based on P/E of 18x our 2024 EPS view, near semiconductor equipment peers. We cut our 2023 EPS view to \$4.69 from \$5.33 and lower our 2024 EPS view to \$6.10 from \$6.32. AEIS prints Q4 adj-EPS of \$1.70 vs. \$1.36, beating the \$1.60 consensus. AEIS realized 23.7% revenue growth Y/Y in Q4 to \$491M, above expectations, driven by ~30% increase in Semi-Equip revenue [-13% Q/Q due to China-related export controls]. Industrial and Medical demand remains very strong, up 21% Y/Y [flat Q/Q], with an uptick in backlog Q/Q [total backlog \$875M, down 20% Q/Q partly due to export controls], despite record shipments with design wins in 3D printing, test/measurements, and lasers [Industrial], as well as surgical, diagnostics, and life sciences equipment [Medical], which we view favorably. Data Center Computing revenue rose 18% Y/Y [+8%] and Telecom and Networking sales increased 15% [+4%]. We expect 2023 growth to be driven by Industrial, Medical, Telecom, and Networking end markets. / Keven Young

November 02, 2022

10:50 AM ET... CFRA Maintains Hold Opinion on Shares of Advanced Energy Industries, Inc. [AEIS 82.91***]:

We cut our 12-month target by \$12 to \$90 based on P/E of 17x our 2023 EPS view, near semi equipment peers. We raise our 2022 EPS to \$6.16 from \$5.20, lower 2023 to \$5.33 from \$6.00, and initiate 2024 EPS at \$6.32. AEIS prints Q3 Adj-EPS of \$2.12 vs. \$0.89, beating the \$1.38 consensus. AEIS realized 49% revenue growth Y/Y in Q3, driven by Semiconductor Equipment sales rising 54% [17% Q/Q] to \$267M, with strength in conductor and dielectric etch. Industrial and Medial revenue increased 48% [14%] to \$120M alongside design win momentum in surgical, life science, and EV charging. Telecom and Networking sales grew 44% [12%] to \$43M, better than expectations. Data Center Computing revenue rose 41% [27%] to \$88M, with several high value design wins in Q3. We see semiconductor equipment revenue declining in 2023, but we like AEIS metal deposition exposure to logic. We are encouraged by AEIS's \$1.1B backlog, with about two-thirds being shippable in the next two quarters. We see FCF of \$80M in 2022 and \$170M in 2023. / Keven Young

August 04, 2022

10:10 AM ET... CFRA Maintains Hold Opinion on Shares of Advanced Energy Industries, Inc. [AEIS 95.62***]:

We lower our 12-month target by \$10 to \$102 based on a P/E of 17x our 2023 EPS view, near peers. We raise our 2022 EPS estimate to \$5.20 from \$5.00 and cut our 2023 EPS view to \$6.00 from \$6.40. AEIS prints Q2 adj-EPS of \$1.44 vs \$1.25, beating the \$1.11 consensus. AEIS was able to grow revenue 22.1% in Q2 to \$441M, beating expectations, driven by semiconductor equipment sales of \$229M [up 30% Y/Y] due to evaluation units for dielectric etch, followed by Industrial & Medical sales of \$105M [+26%] due to factory automation & industrial printing, and Telecom & networking revenue of \$38M [+19%], with design wins in Q2, partially offset by Data Center Computing revenue of \$69M [-9%] negatively impacted by supply constraints. We expect second half 2022 growth to be driven by Industrial & Medical design win momentum and abatement of certain component constraints and continued strength from semiconductor equipment. In addition, we are encouraged by growing backlog [>\$1B in Q2] which is ~80% proprietary products. / Keven Young

May 05, 2022

08:01 AM ET... CFRA Maintains Hold Opinion on Shares of Advanced Energy Industries, Inc. [AEIS 83.46***]:

We raise our 12-month target price by \$2 to \$112, on P/E of 17.5x our 2023 EPS view, a slight discount to semiconductor equipment peers due to lower profitability. We lower our 2022 EPS estimate to \$5.00 from \$5.20 and raise our 2023 EPS view to \$6.40 from \$6.27. AEIS prints Mar-Q adj-EPS of \$1.24 vs. \$1.29, beating the \$0.95 consensus. AEIS posted \$497M in revenue up 13% Y/Y, driven by Data Center Computing [\$76M] grew 29%, Semiconductor Equipment [\$203M] rose 12%, while Industrial & Medical [\$83M] and Telecom & Networking [\$35M] segments which both grew 6%. We expect Q2 sales to be driven by dielectric etch, design win momentum in Industrial & Medical, and 5G infrastructure investments. AEIS completed its acquisition of SL Power Electronics [\$144.5M all cash deal], which we view favorably as it is complementary to AEIS's Industrial & Medical segment, expanding AEIS proforma revenue by ~\$66M and SAM by ~\$400M. We are encouraged by expanding ROIC opportunities post-acquisition [>10% post synergy]. / Keven Young

February 10, 2022

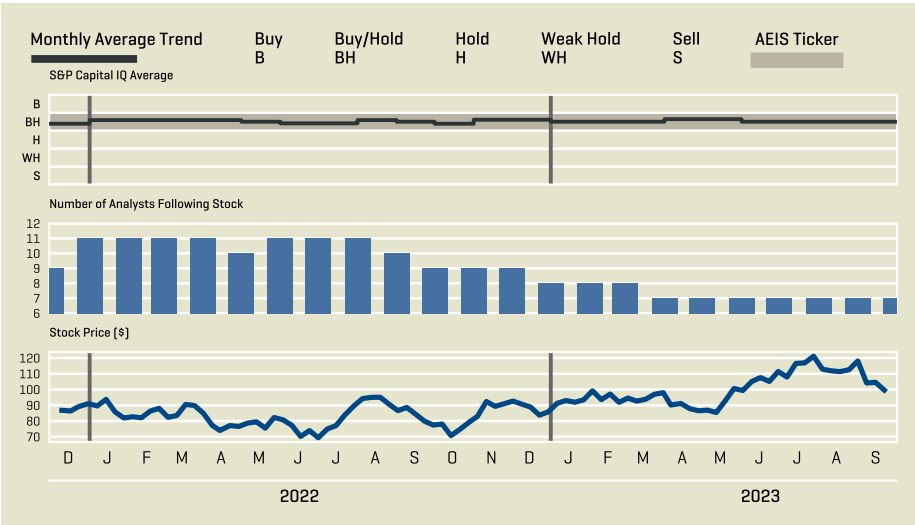
08:18 AM ET... CFRA Maintains Hold Recommendation on Shares of Advanced Energy Industries, Inc [AEIS 85.67***]:

We maintain our 12-month target price of \$110, 17.5x our '23 EPS view, slightly lower than semiconductor equipment peers due to lower profitability. We cut our '22 EPS estimate to \$5.20 from \$6.01 and initiate on our '23 EPS view at \$6.27. AEIS Prints Dec-Q adj-EPS of \$1.36 vs \$1.49, beating the \$0.94 consensus. AEIS posted \$397M in Q4 revenue, a 15% increase sequentially driven by Telecom & Networking which grew 31% QoQ. This growth was followed by Data Center Computer revenue [up 29% QoQ], Industrial & Medical [+22%], and Semiconductor Equipment [+3.4%]. We expect future growth to be driven by Telecom & Networking (T&N) and Industrial & Medical (I&M) markets, as 5G infrastructure ramps continue, 3 new I&M products launch, and design wins from both T&N and I&M begin to come online in 2022. In addition, we are encouraged by several new advanced control capability products for semiconductor equipment but note that we expect supply constraints to remain for 1H '22. / Keven Young

Note: Research notes reflect CFRA's published opinions and analysis on the stock at the time the note was published. The note reflects the views of the equity analyst as of the date and time indicated in the note, and may not reflect CFRA's current view on the company.

Advanced Energy Industries, Inc.

Analysts Recommendations



	No. of Recommendations	% of Total	1 Mo.Prior	3 Mos.Prior
Buy	3	43	3	3
Buy/Hold	1	14	1	1
Hold	3	43	3	3
Weak hold	0	0	0	0
Sell	0	0	0	0
No Opinion	0	0	0	0
Total	7	100	7	7

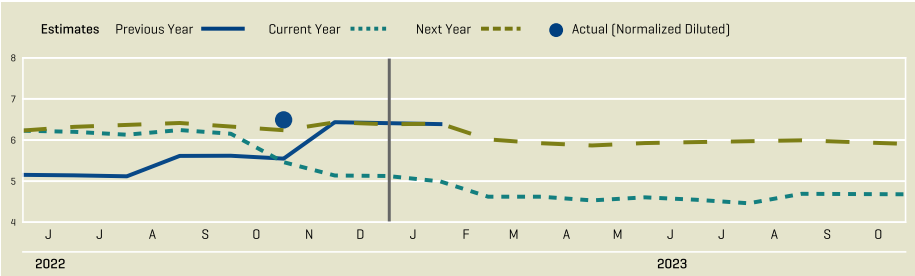
Wall Street Consensus Opinion

Buy/Hold

Wall Street Consensus vs. Performance

For fiscal year 2023, analysts estimate that AEIS will earn USD 4.68. For fiscal year 2024, analysts estimate that AEIS's earnings per share will grow by 26.17% to USD 5.91.

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2024	5.91	6.40	5.57	7	14.97
2023	4.68	4.81	4.51	7	18.89
2024 vs. 2023	▲ 26%	▲ 33%	▲ 23%	N/A%	▼ -21%
Q3'24	1.54	1.60	1.47	4	57.40
Q3'23	1.14	1.18	1.13	7	77.53
Q3'24 vs. Q3'23	▲ 35%	▲ 36%	▲ 30%	▼ -43%	▼ -26%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.
Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

Advanced Energy Industries, Inc.

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index (MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index)], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsize the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share (EPS) estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

- CAGR - Compound Annual Growth Rate
- CAPEX - Capital Expenditures
- CY - Calendar Year
- DCF - Discounted Cash Flow
- DDM - Dividend Discount Model
- EBIT - Earnings Before Interest and Taxes
- EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
- EPS - Earnings Per Share
- EV - Enterprise Value
- FCF - Free Cash Flow
- FFO - Funds From Operations
- FY - Fiscal Year
- P/E - Price/Earnings
- P/NAV - Price to Net Asset Value
- PEG Ratio - P/E-to-Growth Ratio
- PV - Present Value
- R&D - Research & Development
- ROCE - Return on Capital Employed
- ROE Return on Equity
- ROI - Return on Investment
- ROIC - Return on Invested Capital
- ROA - Return on Assets
- SG&A - Selling, General & Administrative Expenses
- SOTP - Sum-of-The-Parts
- WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes (paid in the country of origin).

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



Advanced Energy Industries, Inc.

Disclosures

Stocks are ranked in accordance with the following ranking methodologies:

STARS Stock Reports:

Qualitative STARS rankings are determined and assigned by equity analysts. For reports containing STARS rankings refer to the Glossary section of the report for detailed methodology and the definition of STARS rankings.

Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five [six] model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section seof the report for detailed methodology and the definition of Quantitative rankings.

STARS Stock Reports and Quantitative Stock Reports:

The methodologies used in STARS Stock Reports and Quantitative Stock Reports [collectively, the "Research Reports"] reflect different criteria, assumptions and analytical methods and may have differing rankings. The methodologies and data used to generate the different types of Research Reports are believed by the author and distributor reasonable and appropriate. Generally, CFRA does not generate reports with different ranking methodologies for the same issuer. However, in the event that different methodologies or data are used on the analysis of an issuer, the methodologies may lead to different views on the issuer, which may at times result in contradicting assessments of an issuer. CFRA reserves the right to alter, replace or vary models, methodologies or assumptions from time to time and without notice to clients.

STARS Stock Reports:

Global STARS Distribution as of September 30, 2023

Ranking	North America	Europe	Asia	Global
Buy	36.8%	34.5%	41.6%	37.4%
Hold	53.8%	51.2%	50.3%	52.6%
Sell	9.4%	14.3%	8.1%	10.1%
Total	100.0%	100.0%	100.0%	100.0%

Analyst Certification:

STARS Stock Reports are prepared by the equity research analysts of CFRA and its affiliates and subsidiaries. Quantitative Stock Reports are prepared by CFRA. All of the views expressed in STARS Stock Reports accurately reflect the research analyst's personal views regarding any and all of the subject securities or issuers; all of the views expressed in the Quantitative Stock Reports accurately reflect the output of CFRA's algorithms and programs. Analysts generally update STARS Stock Reports at least four times each year. Quantitative Stock Reports are generally updated weekly. No part of analysts' or CFRA's compensation was, is, or will be directly or indirectly related to the specific rankings or views expressed in any Stock Report.

About CFRA Equity Research:

This Research Report is published and originally distributed by Accounting Research & Analytics, LLC d/b/a CFRA ["CFRA US"], with the following exceptions: In the UK/EU/EEA, it is published and originally distributed by CFRA UK Limited ["CFRA UK"], which is regulated by the Financial Conduct Authority [No. 775151], and in Malaysia by CFRA MY Sdn Bhd [Company No. 683377-A] ["CFRA Malaysia"], which is regulated by Securities Commission Malaysia, [No. CMSL/A0181/2007] under license from CFRA US. These parties and their subsidiaries maintain no responsibility for reports redistributed by third parties such as brokers or financial advisors.

General Disclosure

Notice to all jurisdictions:

Where Research Reports are made available in a language other than English and in the case of inconsistencies between the English and translated versions of a Research Report, the English version will control and supersede any ambiguities between such versions. Neither CFRA nor its affiliates guarantee the accuracy of any translation.

The content of this report and the opinions expressed herein are those of CFRA based upon publicly-available information that CFRA believes to be reliable and the opinions are subject to change without notice. This analysis has not been submitted to, nor received approval from, the United States Securities and Exchange Commission or any other regulatory body. CFRA AND ALL RELATED ENTITIES SPECIFICALLY DISCLAIM ALL WARRANTIES, EXPRESS OR IMPLIED, to the full extent permitted by law, regarding the accuracy, completeness, or usefulness of this information and assumes no liability with respect to the consequences of relying on this information for investment or other purposes.

No content in this Research Report may be modified, reverse engineered, reproduced or distributed in any form by any means, or stored in a database or retrieval system, without the prior written permission of CFRA, or used for any unlawful or unauthorized purposes. Neither CFRA nor its third-party providers, as well as its/their directors, officers, shareholders, employees or agents, guarantee the accuracy, completeness, timeliness or availability of the content herein

Past performance is not necessarily indicative of future results.

This document may contain forward-looking statements or forecasts; such forecasts are

not a reliable indicator of future performance.

This report is not intended to, and does not, constitute an offer or solicitation to buy and sell securities or engage in any investment activity. This report is for informational purposes only. Statements in this report are not made with respect to any particular investor or type of investor. Securities, financial instruments or strategies mentioned herein may not be suitable for all investors and this material is not intended for any specific investor and does not take into account an investor's particular investment objectives, financial situations or needs. Before acting on anything in this report, you should consider whether it is suitable for your particular circumstances and, if necessary, seek professional advice. CFRA may license certain intellectual property or provide services to, or otherwise have a business relationship with, certain issuers of securities that are the subject of CFRA research reports, including exchange-traded investments whose investment objective is to substantially replicate the returns of a proprietary index of CFRA. In cases where CFRA is paid fees that are tied to the amount of assets invested in a fund or the volume of trading activity in a fund, investment in the fund may result in CFRA receiving compensation in addition to the subscription fees or other compensation for services rendered by CFRA, however, no part of CFRA's compensation for services is tied to any particular viewpoint or rating. Additional information on a subject company may be available upon request.

CFRA's financial data provider is S&P Global Market Intelligence. THIS DOCUMENT CONTAINS COPYRIGHTED AND TRADE SECRET MATERIAL DISTRIBUTED UNDER LICENSE FROM S&P GLOBAL MARKET INTELLIGENCE. FOR RECIPIENT'S INTERNAL USE ONLY.

The Global Industry Classification Standard [GICS®] was developed by and/or is the exclusive property of MSCI, Inc. and S&P Global Market Intelligence. GICS is a service mark of MSCI and S&P Global Market Intelligence and has been licensed for use by CFRA.

Other Disclaimers and Notices

Certain information in this report is provided by S&P Global, Inc. and/or its affiliates and subsidiaries [collectively "S&P Global"]. Such information is subject to the following disclaimers and notices: "Copyright © 2018, S&P Global Market Intelligence [and its affiliates as applicable]. All rights reserved. Nothing contained herein is investment advice and a reference to a particular investment or security, a credit rating or any observation concerning a security or investment provided by S&P Global is not a recommendation to buy, sell or hold such investment or security or make any other investment decisions. This may contain information obtained from third parties, including ratings from credit ratings agencies. Reproduction and distribution of S&P Global's information and third party content in any form is prohibited except with the prior written permission of S&P Global or the related third party, as applicable. Neither S&P Global nor its third party providers guarantee the accuracy, completeness, timeliness or availability of any information, including ratings, and are not responsible for any errors or omissions [negligent or otherwise], regardless of the cause, or for the results obtained from the use of such information or content. S&P GLOBAL AND ITS THIRD PARTY CONTENT PROVIDERS GIVE NO EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, ANY WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE AND ALL S&P INFORMATION IS PROVIDED ON AN AS-IS BASIS. S&P GLOBAL AND ITS THIRD PARTY CONTENT PROVIDERS SHALL NOT BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, EXEMPLARY, COMPENSATORY, PUNITIVE, SPECIAL OR CONSEQUENTIAL DAMAGES, COSTS, EXPENSES, LEGAL FEES, OR LOSSES [INCLUDING LOST INCOME OR PROFITS AND OPPORTUNITY COSTS OR LOSSES CAUSED BY NEGLIGENCE] IN CONNECTION WITH ANY USE OF THEIR INFORMATION OR CONTENT, INCLUDING RATINGS. Credit ratings are statements of opinions and are not statements of fact or recommendations to purchase, hold or sell securities. They do not address the suitability of securities or the suitability of securities for investment purposes, and should not be relied on as investment advice."

Advanced Energy Industries, Inc.

CFRA's Research Reports may be distributed in certain localities, countries and/or jurisdictions by independent third parties or independent intermediaries and/or distributors ("Intermediaries"). Intermediaries are not acting as agents or representatives of CFRA. In territories where an Intermediary distributes CFRA's Research Reports, the Intermediary, and not CFRA, is solely responsible for complying with all applicable regulations, laws, rules, circulars, codes and guidelines established by local and/or regional regulatory authorities, including laws in connection with the distribution of third party research reports, licensing requirements, supervisory and record keeping obligations that the Intermediary may have under the applicable laws and regulations of the territories where it distributes the Research Reports.

For residents of the European Union/European Economic Area:

Research reports are originally distributed by CFRA UK Limited (company number 08456139 registered in England & Wales with its registered office address at New Derwent House, 69-73 Theobalds Road, London, WC1X 8TA, United Kingdom). CFRA UK Limited is regulated by the UK Financial Conduct Authority (No. 775151).

For residents of Malaysia:

Research reports are originally produced and distributed by CFRA MY Sdn Bhd (Company No. 683377-A) ("CFRA Malaysia"), a wholly-owned subsidiary of CFRA US. CFRA Malaysia is regulated by Securities Commission Malaysia (License No. CMSL/A0181/2007).

For Recipients in Canada:

This report is not prepared subject to Canadian disclosure requirements and may not be suitable for Canadian investors.

For residents of Singapore:

This Research Report is distributed by CFRA UK Limited to its clients in Singapore who hold a financial advisers licence or is a person exempt from holding such licence ("SG Intermediary"). Recipients of this Research Report in Singapore should contact the SG Intermediary in respect to any matters arising from, or in connection with, the analysis in this report. Where the recipient is not an accredited, expert or institutional investor as defined by the Securities and Futures Act, the SG Intermediary accepts legal responsibility for the contents of this Research Report in accordance with applicable law. When reports are distributed by SG Intermediaries in Singapore, the SG Intermediary, and not CFRA, is solely responsible for ensuring that the recipients of the Research Reports understand the information contained in the Research Reports and that such information is suitable based on the customer's profile and investment objectives. This Research Report is intended for general circulation and no advice or recommendation is made herein or by CFRA to any particular person. CFRA does not assume any responsibility to advise on whether any particular product is suitable for any person, and the analysis herein does not take into account the specific investment objectives, financial situation or particular needs of any particular person, and should not be relied upon for any investment decision.

For residents of all other countries:

Research reports are originally distributed Accounting Research & Analytics, LLC d/b/a CFRA.

Copyright © 2023 CFRA. All rights reserved. CFRA and STARS are registered trademarks of CFRA.

EFG Disclaimer

The attached document is produced for it to be used by entities, including affiliates and branches, of EFG International AG. The information herein is provided on the basis that it is a suitable investment for the recipient taking into account the recipient's personal circumstances and should therefore be considered as a personal recommendation. Tax implications should be nevertheless independently assessed by the recipient with a tax advisor before entering into any transaction.

The information provided herein is for the exclusive use of the recipient and may not be reproduced, disclosed or distributed, neither in part nor in full.

Investment products may be subject to investment risks involving, but not limited to, possible loss of all or part of the principal invested.

The information and views expressed herein at the time of writing are subject to change at any time without notice and there is no obligation to update or remove outdated information. Historical data on the performance of the securities and financial instruments or the underlying assets in this document is no indication for future performance and the value of investments may fall as well as rise. This document is not directed at, or intended for distribution to or use by, any person or entity domiciled or resident in any jurisdiction where such distribution, publication, availability or use would be contrary to applicable laws or regulations of such jurisdictions.

EFG and its employees may engage in securities transactions, on a proprietary basis or otherwise and hold long or short positions with regard to the instruments identified herein; such transactions or positions may be inconsistent with the views expressed in the document.

If you have received this document from any affiliate or branch referred to below, please note the following:

Switzerland: EFG Bank AG, Zurich, including its Geneva and Lugano branches, is authorised and regulated by the FINMA. Registered Office: EFG Bank AG, Bleicherweg 8, 8001 Zurich, Switzerland. Registered Swiss Branches: EFG Bank SA, 24 quai du Seujet, 1211 Geneva 2, and EFG Bank SA, Via Magatti 2, 6900 Lugano.

The information provided in this document is not the result of a financial research conducted by the Bank's research department. Therefore, the "Directive on the Independence of Financial Research" issued by the Swiss Banking Association does not apply to it. As the document is not intended to be exhaustive on all risks related to financial instruments, the recipient should consult the specific product documentation and the June 2023 "Risks Involved in Trading Financial Instruments" brochure released by the Swiss Bankers Association and available on the Swiss Bankers Association website:

<https://www.swissbanking.org/en/services/library/guidelines> or free of charge from EFG Bank AG.

Luxembourg: EFG Bank (Luxembourg) S.A. is authorised by the Ministry of Finance Luxembourg and supervised by the Commission de Surveillance du Secteur Financier (CSSF). EFG Bank (Luxembourg) S.A. is Member of the Deposit Guarantee Fund Luxembourg (F.G.D.L. - Fonds de Garantie des Dépôts Luxembourg) and Member of the Luxembourg Investor Compensation Scheme (S.I.I.L. - Système d'Indemnisation des Investisseurs Luxembourg). R.C.S. Luxembourg no. B113375. Registered address: EFG Bank (Luxembourg) S.A. - 56, Grand-Rue, L-1660 Luxembourg.

Portugal: EFG Bank (Luxembourg) S.A. - Sucursal em Portugal is authorised and supervised by Banco de Portugal (register 280) and the CMVM, the Portuguese securities market commission, (register 393) for the provision of financial advisory and reception and transmission of orders. EFG Bank (Luxembourg) S.A. - Sucursal em Portugal is a non-booking branch of EFG Bank (Luxembourg) S.A., a public limited liability company incorporated under the laws of the Grand Duchy of Luxembourg, authorised and supervised by the CSSF (Commission de Surveillance du Secteur Financier). Lisbon Head Office: Avenida da Liberdade n.º 131 - 6º Dto., 1250 - 140 Lisboa. Porto agency: Avenida da Boavista, n.º 1837 - Escritório 6.2, 4100-133 Porto. Companies Registry Number: 980649439.

Greece: EFG Bank (Luxembourg) S.A., Athens Branch is an non-booking establishment of EFG Bank (Luxembourg) S.A. which is authorised to promote EFG Bank (Luxembourg) S.A.'s Greece: EFG Bank (Luxembourg) S.A., Athens Branch is an non-booking establishment of EFG Bank (Luxembourg) S.A. which is authorised to promote EFG Bank (Luxembourg) S.A.'s products and services based on the EU freedom of establishment pursuant to a license granted by the Luxembourg financial supervisory authority "CSSF". Registered address: 342 Kifisias Ave. & Ethnikis Antistaseos Str. - 154 51 N. Psychiko, General Commercial Registry no. 143057760001.

Liechtenstein: EFG Bank von Ernst AG is regulated by the Financial Market Authority Liechtenstein. Registered address: EFG Bank von Ernst AG Egertastrasse 10 - 9490 Vaduz, Liechtenstein.

Monaco: EFG Bank (Monaco) SAM is a Monegasque Public Limited Company with a company registration no. 90 S 02647 (Registre du Commerce et de l'Industrie de la Principauté de Monaco). EFG Bank (Monaco) SAM is a bank with financial activities authorised and regulated by the French Prudential Supervision and Resolution Authority and by the Monegasque Commission for the Control of Financial Activities. Registered address: EFG Bank (Monaco) SAM, Villa les Aigles, 15, avenue d'Ostende - BP 37 - 98001 Monaco (Principauté de Monaco), telephone: +377 93 15 11 11. The recipient of this document is perfectly fluent in English and waives the possibility to obtain a French version of this publication.

United Kingdom: EFG Private Bank Limited is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority. EFG Private Bank Limited is a member of the London Stock Exchange. Registered company no. 2321802. Registered address: EFG Private Bank Limited, Park House, 116 Park Street, London W1K 6AP, United Kingdom, telephone +44 (0)20 7491 9111.

Dubai: EFG (Middle East) Limited is regulated by the DFSA. This material is intended "for professional clients only". Registered address: EFG (Middle East) Limited DIFC, Gate Precinct 5, 7th Floor PO Box 507245 - Dubai, UAE.